Salem Yousry Abdelaziz

AI Automation Engineer | LLM Applications | Business Strategy (MBA)

Cairo, Egypt | +20 151 536 3773 | salemyousry4@gmail.com | linkedin.com/in/salem-yousry-28655337b | github.com/salemyousry4-design

PROFESSIONAL SUMMARY

MBA-qualified business and strategy consultant transitioning into AI engineering, with 4 years of business development experience in the restaurant and F&B sector. Builds Python applications end to end, from client discovery and requirements through APIs, real-time web interfaces, automated tests and retrieval-augmented AI assistants. Grounded in machine learning (certified, 2026), with six published ML projects. Brings the commercial judgment to frame a messy business problem, design a practical solution, and explain it to non-technical decision-makers. Works in Arabic and English; open to remote roles with European teams.

PROJECTS

Sky Sync AI: Offline Fleet Maintenance & Readiness System — 2026

Python · FastAPI · WebSockets · SQLite · RAG (BM25) · local LLM (Ollama) · Docker · CI · live demo

- Led product discovery for a helicopter fleet operator whose paper logs let aircraft fly while overdue for inspection; defined measurable success criteria and the MVP scope.
- Built a fully offline (air-gapped) system on a local network: FastAPI server with live WebSocket sync, so every station (commander, tower, engineers, pilots) sees the same fleet state in real time.
- Role-based access for six user types with four-eyes sign-off (the engineer records the work, the chief engineer certifies it), TOTP two-factor login, account lockout and an append-only activity log.
- Explainable 0–100 aircraft health score; overdue aircraft (50-hour / 30-day rule) are blocked from takeoff automatically.
- AI maintenance assistant with retrieval-augmented generation: BM25 search over the maintenance library plus live fleet data, answered by a local LLM that must cite its sources and never guess.
- 62 automated tests; Docker packaging, CI pipeline and a one-click Windows installer.

Menem Bot: AI Ordering Kiosk for a Grill Restaurant — In progress · 2026

Python · LLMs (Ollama / cloud) · JSON

- Scoping an in-store AI assistant for a real restaurant client that takes orders in Egyptian Arabic and suggests add-ons.
- Structured the full menu (39 items, size-based pricing) as JSON; the bill is computed deterministically in Python, never by the LLM, to guarantee correct totals.

Machine Learning Portfolio — 2026

scikit-learn · XGBoost · TensorFlow / Keras · Streamlit · Plotly · github.com/salemyousry4-design

- **Breast cancer classification:** 4-model comparison with 10-fold CV; XGBoost reaches 97.1% accuracy with 100% recall on malignant cases (live app).
- **Cat vs dog image classifier:** CNN from scratch vs MobileNetV2 transfer learning; 98% accuracy on 2,000 unseen images.
- **Bank churn prediction:** neural network on 10,000 customers, 85.7% accuracy / 0.856 ROC-AUC, with leakage-safe preprocessing.
- **Purchase prediction & dimensionality reduction:** tuned RBF-kernel SVM (91.7% CV accuracy); LDA reduces 13 features to 2 at 99.3% CV accuracy.

CORE SKILLS

AI / LLM	Retrieval-augmented generation (BM25, embeddings), prompt engineering, tool / function calling, structured outputs, local LLMs (Ollama)
Software	Python (OOP, packages, custom exceptions), FastAPI, WebSockets, SQLite / SQL, pytest, Docker, Git & GitHub Actions, technical documentation
Machine Learning	Regression, classification, clustering, NLP, deep learning (CNNs, transfer learning), PCA / LDA, model selection & XGBoost; NumPy, pandas, scikit-learn, TensorFlow
Business	Discovery & stakeholder interviews, go-to-market strategy, feasibility studies, financial modelling, B2B partnerships, contract negotiation
Languages	Arabic (native), English (fluent), German (elementary, A2, actively studying)

PROFESSIONAL EXPERIENCE

Business Development Consultant — 2022 – Present

Restaurant & F&B Sector · Cairo, Egypt

- Automated business reporting and operational forecasting workflows, cutting manual analysis cycle time by 40%.
- Negotiated B2B agreements and built strategic alliances with regional technology providers and operational partners.
- Shaped go-to-market strategies with data analysis and market research; led engagements end to end, from scoping the commercial problem to presenting feasibility studies and financial models to senior management.

CERTIFICATION & EDUCATION

Machine Learning A-Z: ML, DL, AI with AWS, Python & R — Oct 2026

Udemy, SuperDataScience · 49.5 hours · Credential UC-4a4fad90-7b21-4dd7-b2ca-e6ed757c9feb

Master of Business Administration (MBA), Technology Management & Strategic Business Growth — 2023

Arab Academy for Science, Technology and Maritime Transport (AASTMT) · Cairo, Egypt